Applications of Information and Communication Technologies

Lecture-5

Today's Agenda

- Types of storage devices,
- Characteristics of different storage devices

Storage

- A storage medium is the physical material on which a computer keeps data, information, programs, and applications
- Cloud storage keeps information on servers on the Internet, and the actual media on which the files are stored are transparent to the user

Storage


storage
internal hard disk
internal solid-state drive
SSD
1TB
external hard drive
memory cards
USB flash drive
cloud storage
optical discs
network attached
storage device
magnetic
stripe card
smart card
RFID tags
NFC tags

Storage

- Capacity is the number of bytes a storage medium can hold.

Table 8-1 Terms Used to Define Storage

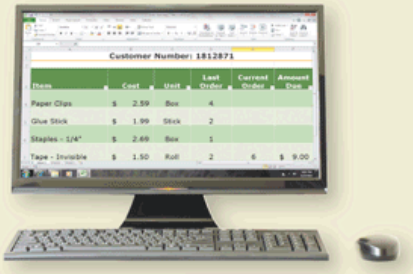
Storage Term	Approximate Number of Bytes	Exact Number of Bytes
Kilobyte (KB)	1 thousand	2^{10} or 1,024
Megabyte (MB)	1 million	2^{20} or 1,048,576
Gigabyte (GB)	1 billion	2^{30} or 1,073,741,824
Terabyte (TB)	1 trillion	2^{40} or 1,099,511,627,776
Petabyte (PB)	1 quadrillion	2^{50} or 1,125,899,906,842,624
Exabyte (EB)	1 quintillion	2^{60} or 1,152,921,504,606,846,976
Zettabyte (ZB)	1 sextillion	2^{70} or 1,180,591,620,717,411,303,424
Yottabyte (YB)	1 septillion	2^{80} or 1,208,925,819,614,629,174,706,176

Storage versus Memory

- Items on a storage medium remain intact even when you turn off a computer or mobile device. Thus, a storage medium is non-volatile.
- Most memory (i.e., RAM) holds data and instructions temporarily and, thus, is volatile. Figure 8-2 illustrates this concept of volatility.

Storage versus Memory (Figure 8-2)

An Illustration of Volatility

State of Computer	Screen Display Volatile	Contents of Most RAM Volatile	Contents of Storage Nonvolatile
 ON		 invisible tape $1.50 per roll 6 rolls $9.00 total due	 paper clips $2.59 per box glue stick $1.99 per stick 1/4" staples $2.69 per box invisible tape $1.50 per roll
 OFF			 paper clips $2.59 per box glue stick $1.99 per stick 1/4" staples $2.69 per box invisible tape $1.50 per roll

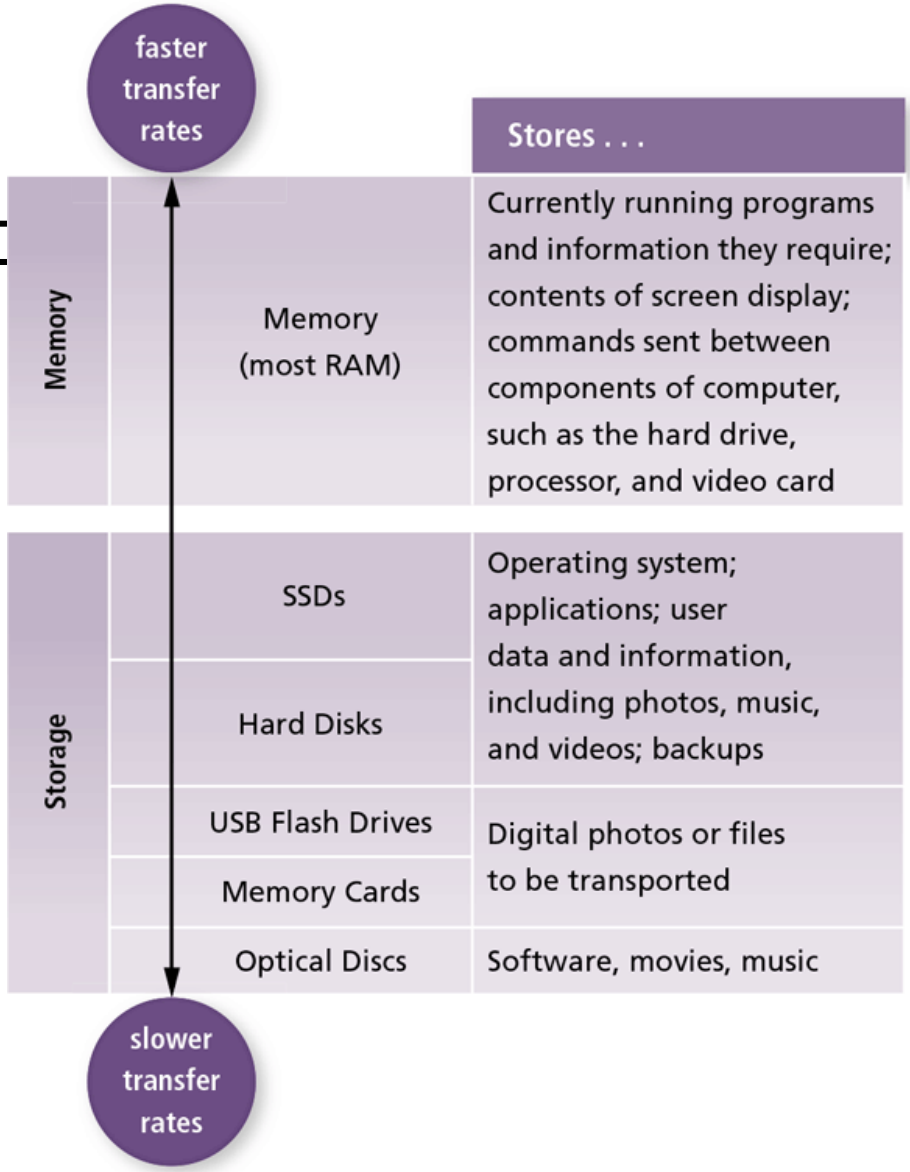
How do storage and memory interact?

- When you turn on a computer or mobile device, it locates the operating system on its storage medium and loads the operating system into its memory (specifically, RAM).
- When you issue a command to run an application, such as a browser, the operating system locates the application on a storage medium and loads it into memory (RAM).
- When you are finished using the application, the operating system removes it from RAM, but the application remains on the storage medium.

Access time

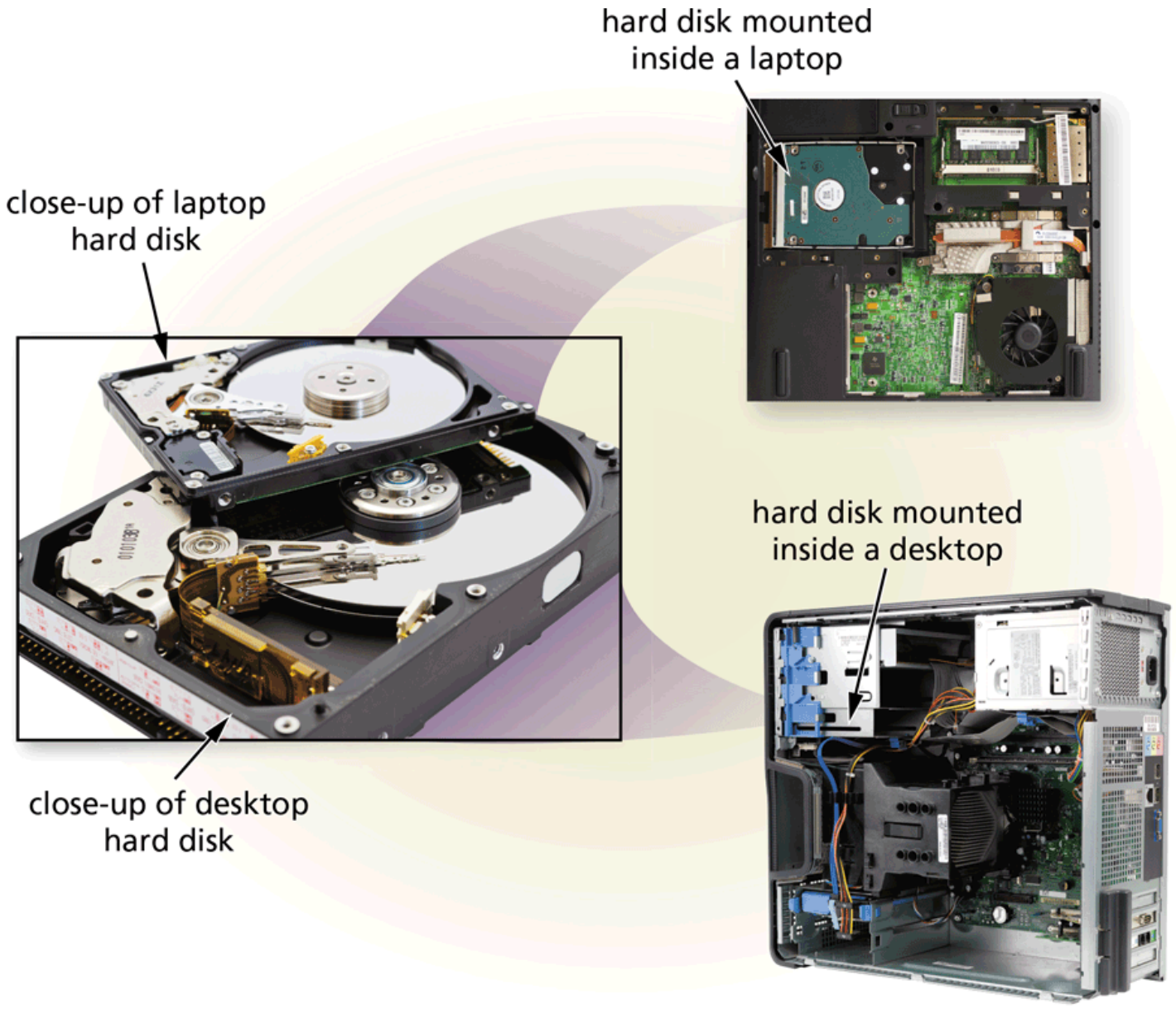
Access time measures:

- The amount of time it takes a storage device to locate an item on a storage medium
- The time required to deliver an item from memory to the processor


faster transfer rates
Stores . . .
Memory
Memory (most RAM)
Currently running programs and information they require; contents of screen display; commands sent between components of computer, such as the hard drive, processor, and video card
Storage
SSDs
Operating system; applications; user data and information, including photos, music, and videos; backups
Hard Disks
USB Flash Drives
Digital photos or files to be transported
Memory Cards
Optical Discs
Software, movies, music
slower transfer rates

Hard Drives

- A hard disk, also called a hard disk drive (HDD) contains one or more fixed, circular platters that use magnetic particles to store data, instructions, and information


hard disk mounted
inside a laptop
close-up of laptop
hard disk
hard disk mounted
inside a desktop
close-up of desktop
hard disk

Storage Capacity Of Hard Disks

- The storage capacity of hard disks varies and is determined by:
  1. The number of platters the hard disk contains
  2. Whether the disk uses longitudinal or perpendicular recording
  3. Density

1. Platter

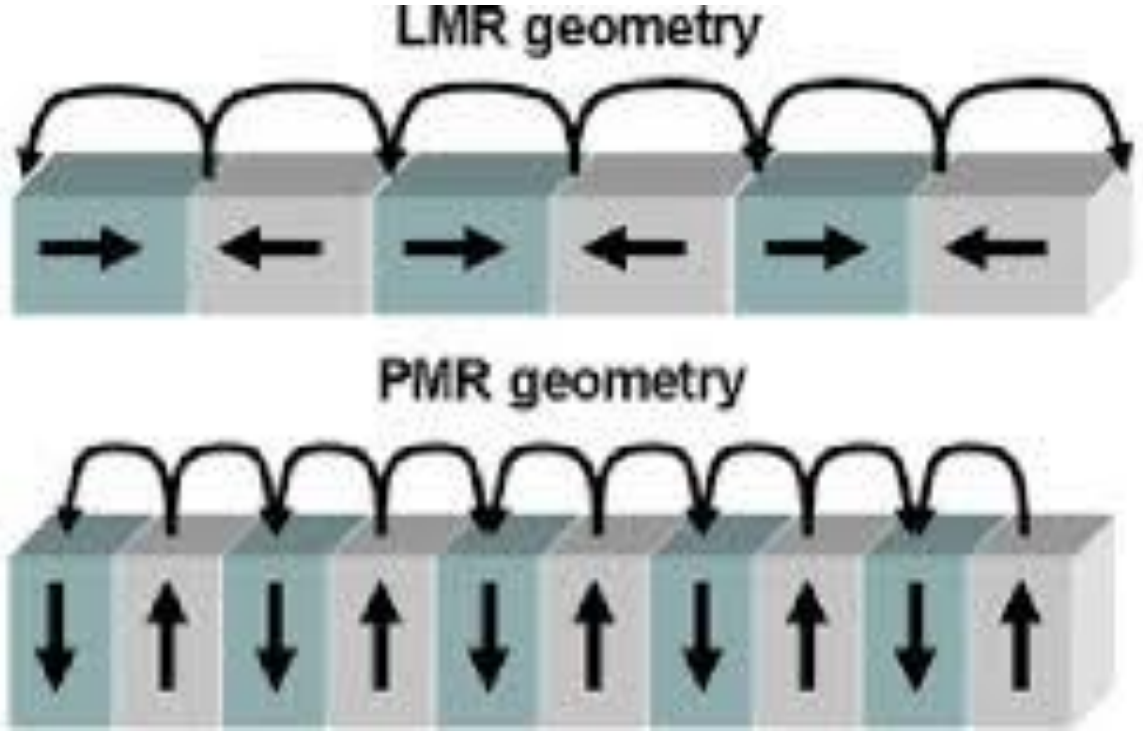
- A platter is made of aluminum, glass, or ceramic and has a thin coating of alloy material that allows items to be recorded magnetically on its surface.



2. Longitudinal Or Perpendicular Recording

- Longitudinal recording aligns the magnetic particles horizontally around the surface of the disk.
- With perpendicular recording, by contrast, hard disks align the magnetic particles vertically, or perpendicular to the disk's surface making much storage capacity possible

LMR vs PMR

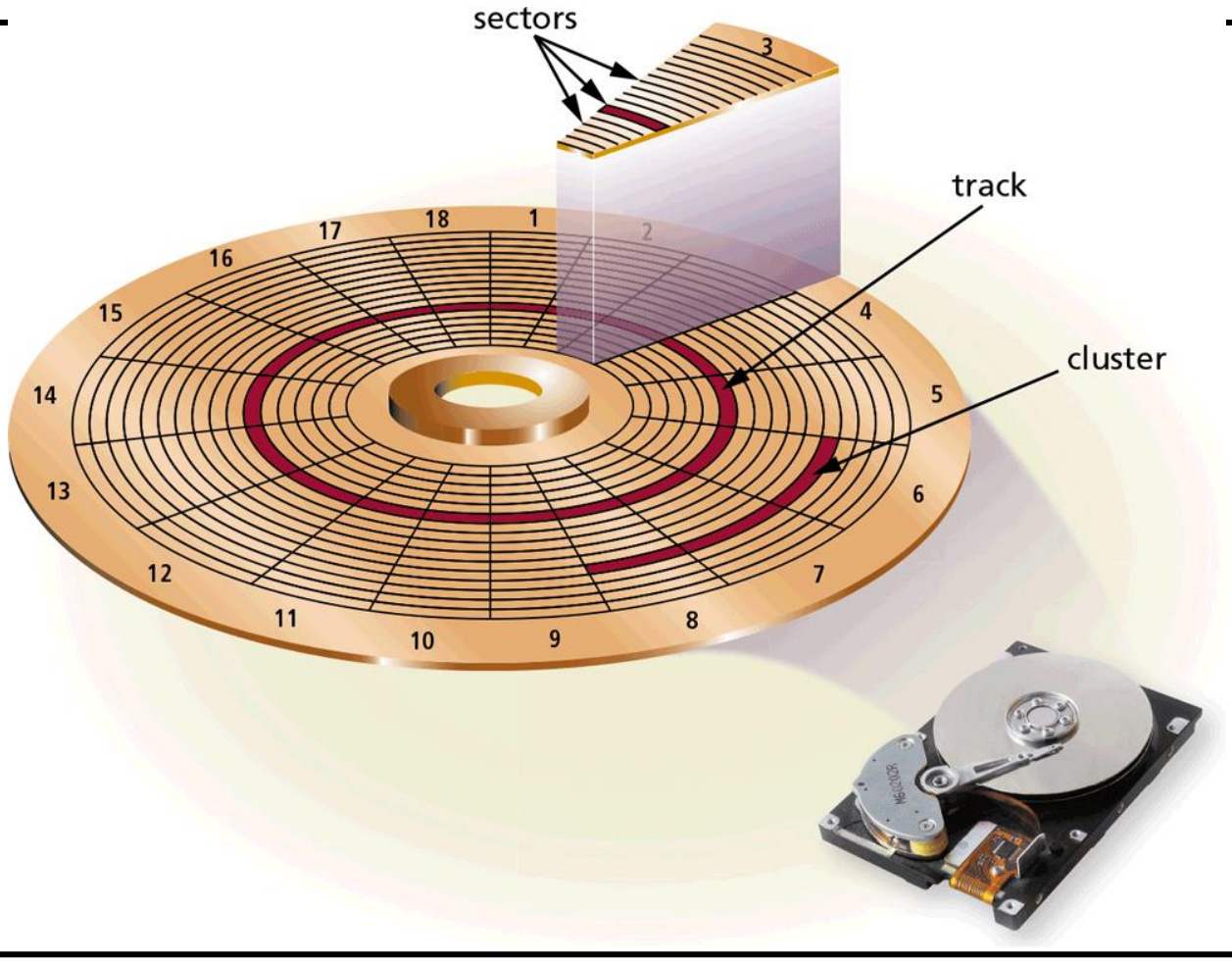


3. Density

- Density is the number of bits in an area on a storage medium.
- A higher density means more storage capacity.

Formatting

- Hard disks are read/write storage media.
- Before any data can be read from or written on a hard disk, the disk must be formatted.
- Formatting is the process of dividing the disk into tracks and sectors so that the operating system can store and locate data and information on the disk.


sectors
track
cluster
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18

Tracks and Sectors

- Tracks form circles on the surface of a hard disk.
- A track is a narrow recording band that forms a full circle on the surface of the disk.

Sectors

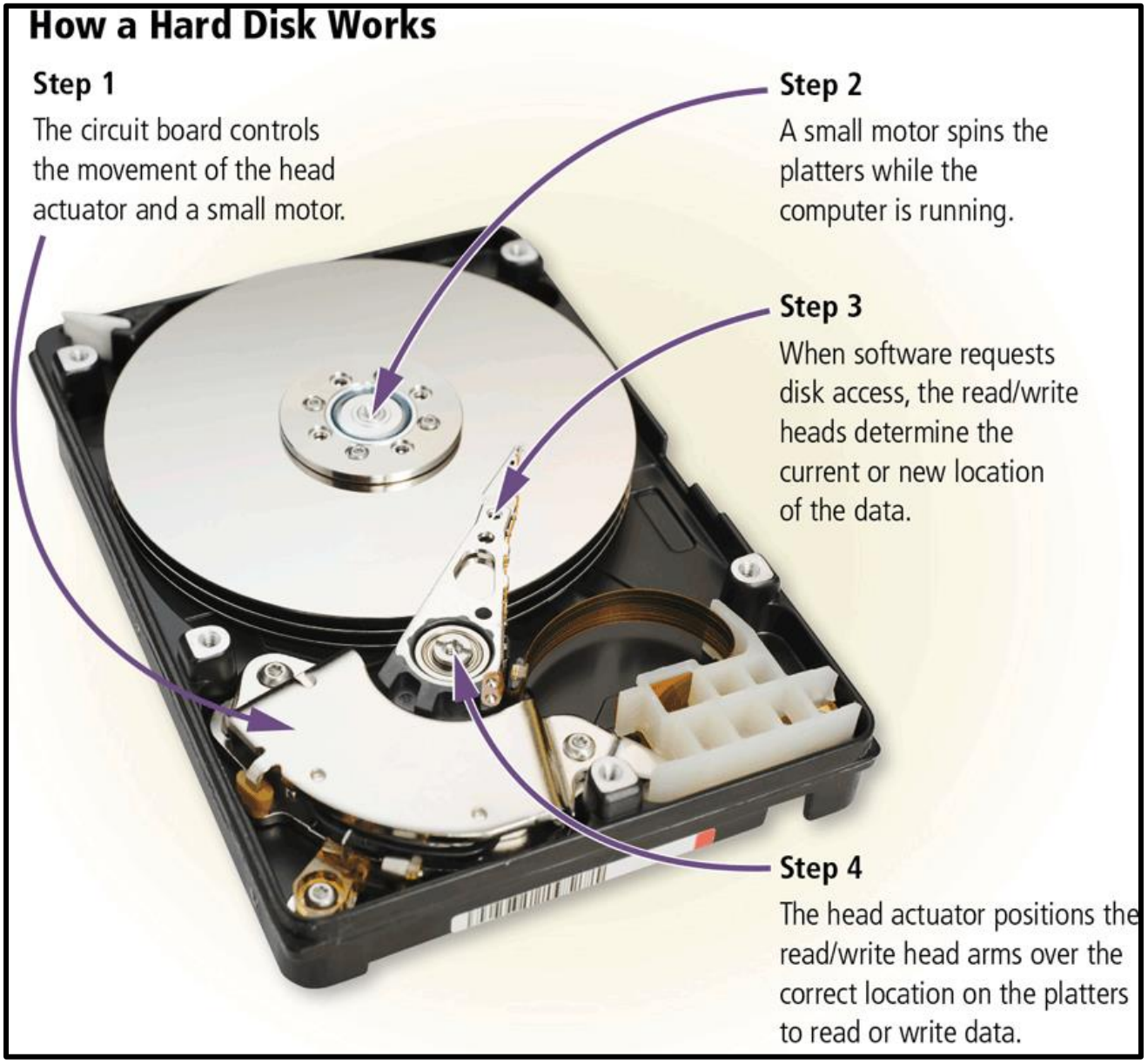
- The disk's storage locations consist of wedge-shaped sections, which break the tracks into small arcs called sectors.
- Several sectors form a cluster.
- A sector typically stores up to 512 bytes of data.
- Sometimes, a sector has a flaw and cannot store data. When you format a disk, the operating system marks these bad sectors as unusable.

Form Factor

- On desktops, the platters most often have a form factor (size) of approximately 3.5 inches in diameter.
- On laptops, mobile devices, and some servers, the form factor is 2.5 inches or less.

Read/Write Heads

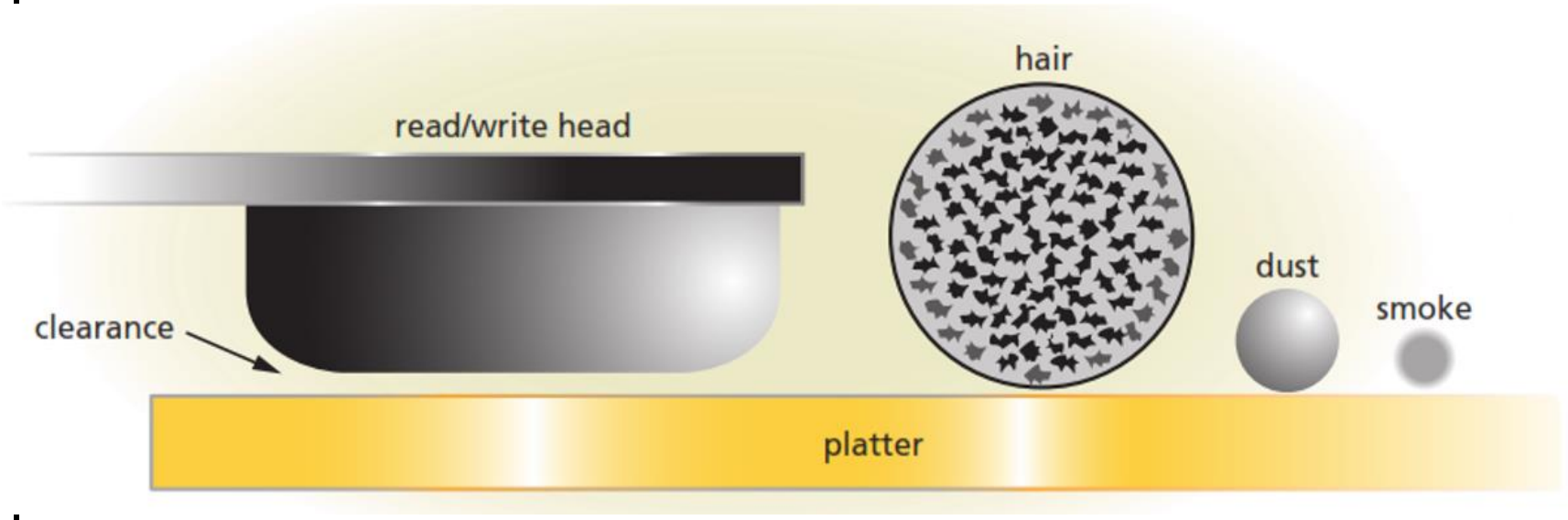
- A typical hard disk has multiple platters stacked on top of one another.
- Each platter has two read/write heads, one for each side.
- A read/write head is the mechanism that reads items and writes items in the drive as it barely touches the disk's recording surface. A head actuator on the hard disk attaches to arms that move the read/write heads to the proper location on the platter.


How a Hard Disk Works
Step 1
The circuit board controls the movement of the head actuator and a small motor.
Step 2
A small motor spins the platters while the computer is running.
Step 3
When software requests disk access, the read/write heads determine the current or new location of the data.
Step 4
The head actuator positions the read/write head arms over the correct location on the platters to read or write data.

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Head Crash



Head Crash

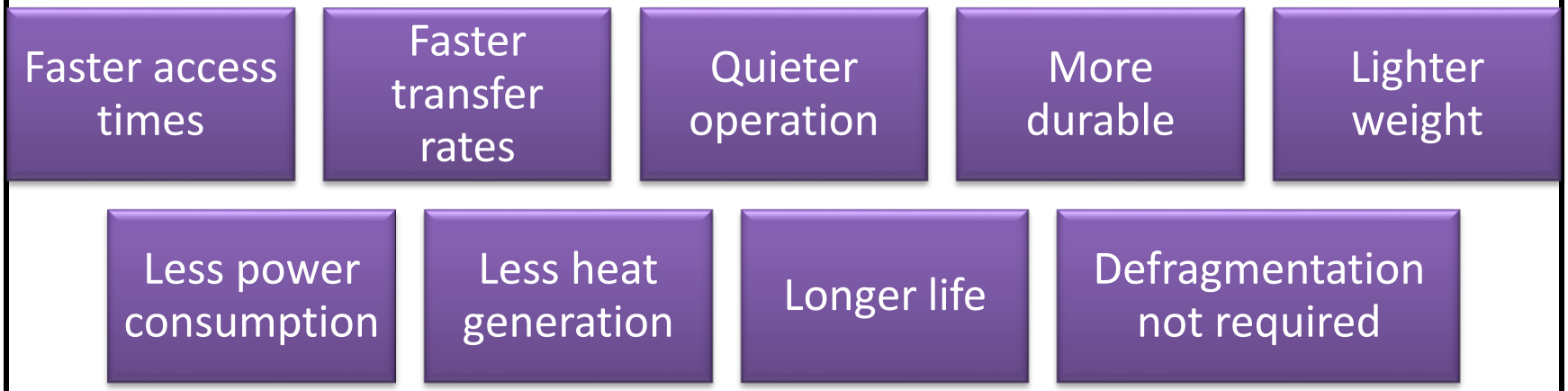
- What happens if dust touches the surface of a platter on a hard disk?
- Because of the close clearance between the read/write head and the platter on a hard disk, dust, dirt, hair, smoke, or any other contaminant could cause a disk to crash (Figure 8-7). A head crash occurs when a read/write head touches the surface of a platter, usually resulting in a loss of data or sometimes loss of the entire disk.

Defragmentation

- Defragmenting a hard disk can improve your computer's performance by storing all related files for a particular program together.
- This reduces the amount of time it takes the hard disk to locate and access the files necessary for programs to run.
- Windows has a built-in tool to defragment a computer's hard disk. Schedule regular defragmenting to optimize performance of your computer disks
- Because the Mac OS defragments files automatically and writes smaller files closer together, Mac users generally do not have to defragment their hard disk.

SSD (Solid State Drive)

- An SSD (solid state drive) is a flash memory storage device that contains its own processor to manage its storage
- Form factor: 3.5 inches, 2.5 inches, 1.8 inches



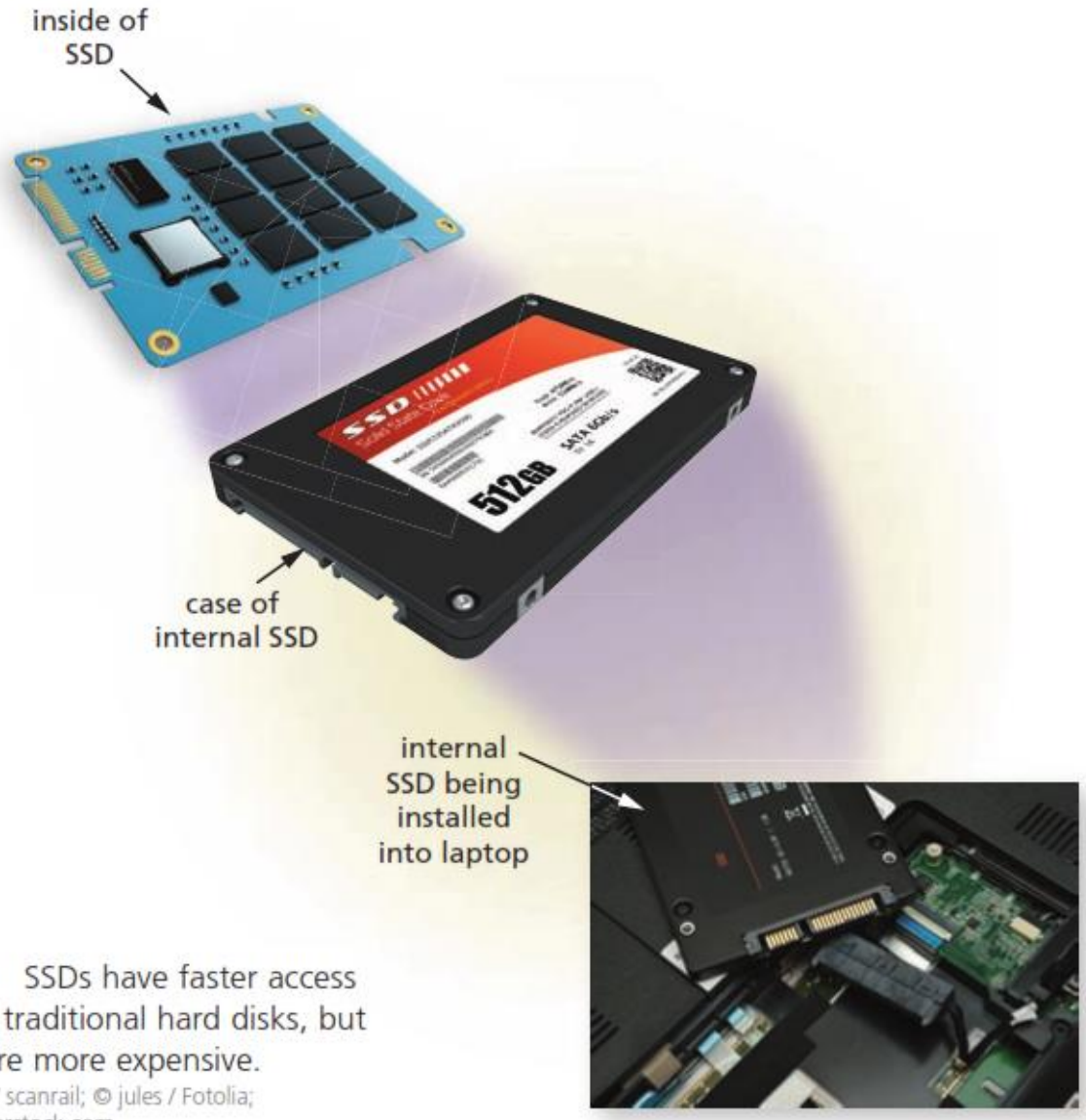


Figure 8-8 SSDs have faster access times than traditional hard disks, but they also are more expensive.

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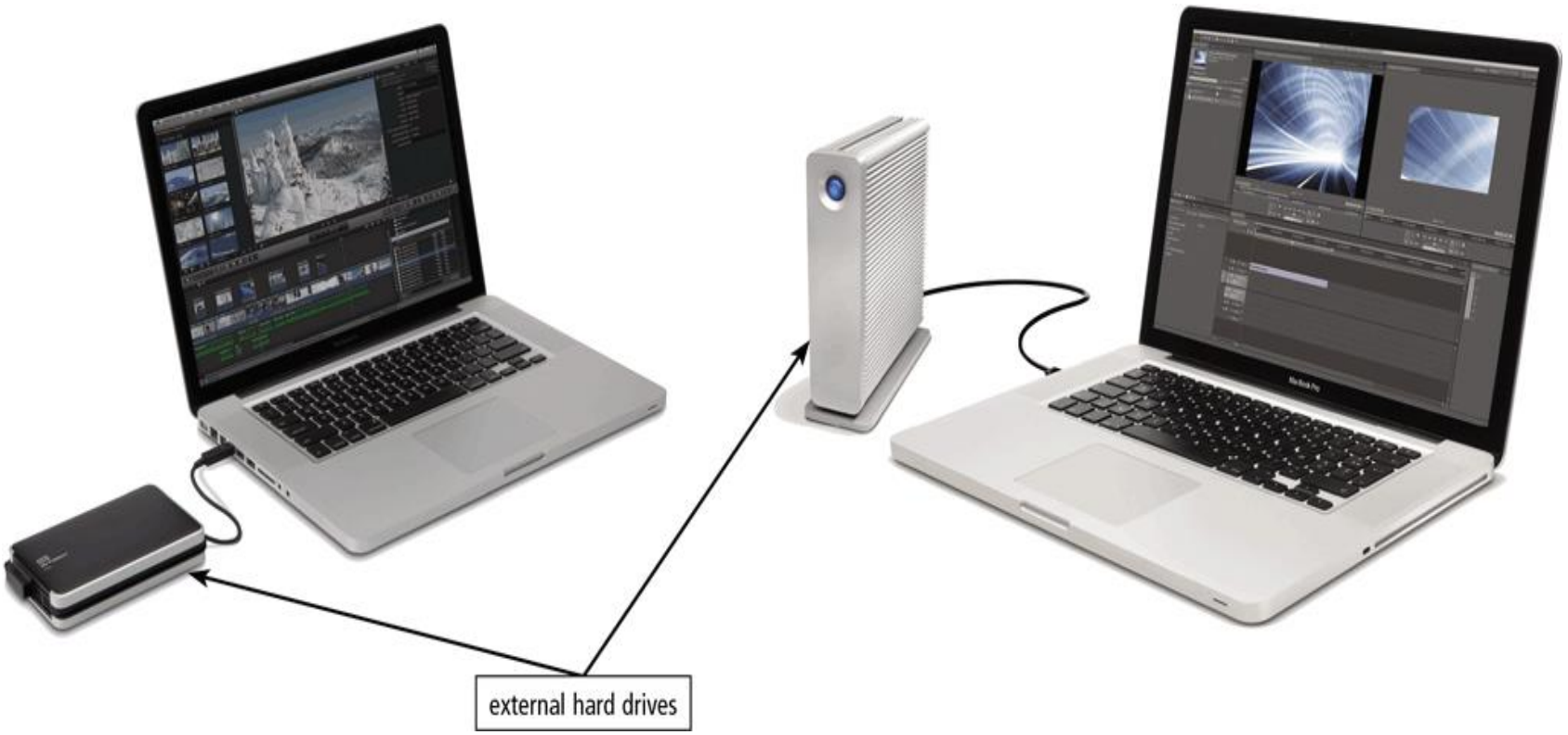
Which should you use, a magnetic hard disk or SSD?

- Magnetic hard disk if:
  - lowest-cost option,
  - basic tasks at one location,
  - require a large amount of storage space
- SSD If:
  - portability,
  - want faster access to stored items,
  - need a quieter drive.
- Dual-drive computer: includes both a hard disk and SSD, benefits of both drives.

External Hard Disk

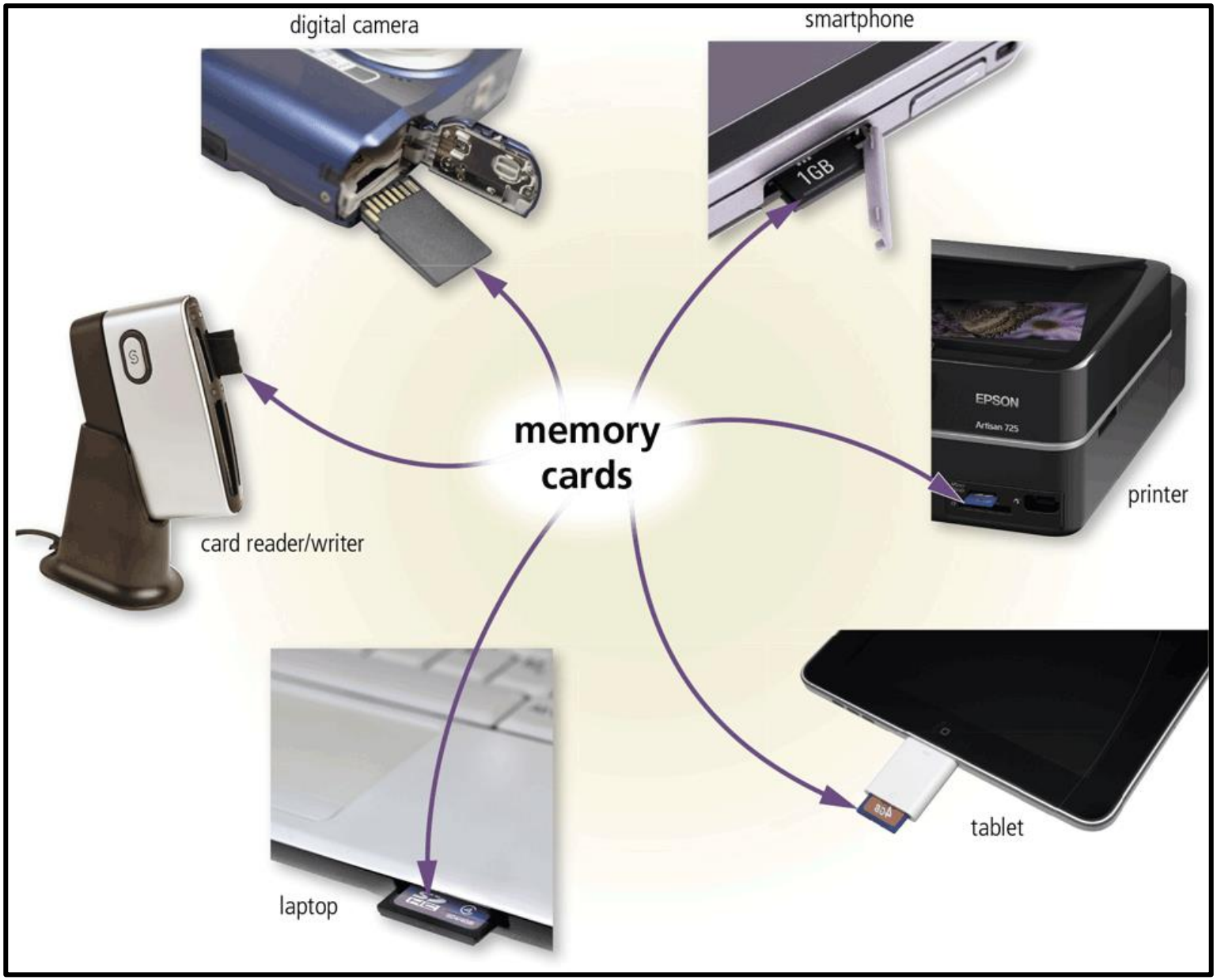
- An external hard disk is a separate freestanding storage device that connects with a cable to a USB port or other port on a computer or mobile device

External Hard Disk



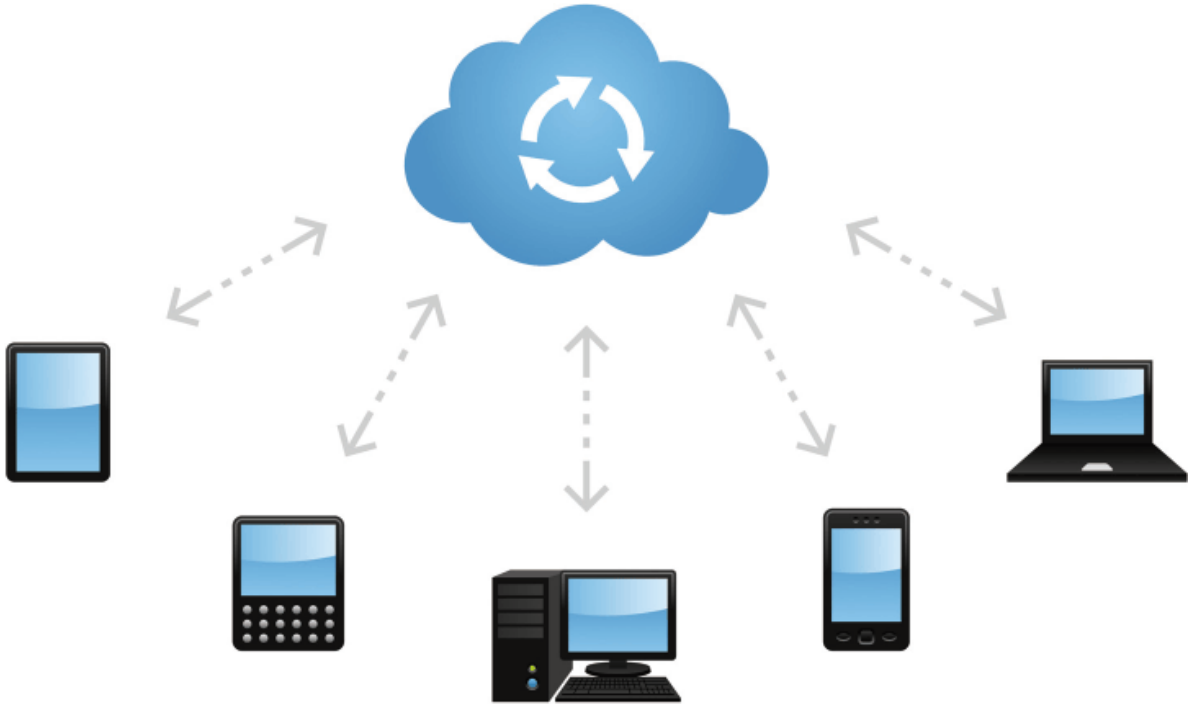
Portable Flash Memory Storage

- A memory card is a removable flash memory storage device that you insert and remove from a slot in a computer, mobile device, or card reader/writer
- USB flash drives plug into a USB port on a computer or mobile device


digital camera
smartphone
1GB
EPSON
Artisan 725
printer
memory cards
card reader/writer
laptop
tablet

Cloud Storage

- Cloud storage is an Internet service that provides storage to computer or mobile device users



Advantages Of Cloud Storage

- To access files on the Internet from any computer or device that has Internet access
- To store large audio, video, and graphics files on the Internet instantaneously
- To allow others to access their files on the Internet
- To view time-critical data and images immediately while away from the main office or location; for example, doctors can view X-ray images from another hospital, home, or office, or while on vacation
- To store off-site backups of data
- To provide data center functions, relieving enterprises of this task

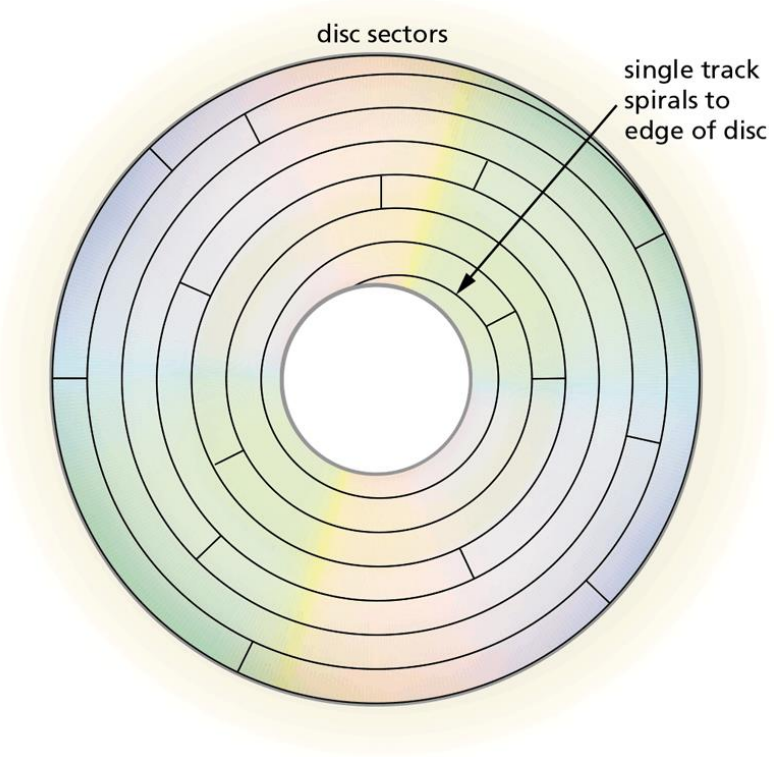
Optical Discs

- An optical disc consists of a flat, round, portable disc made of metal, plastic, and lacquer that is written and read by a laser



Optical Discs

- Optical discs store items in a single track that spirals from the center of the disc to the edge of the disc
- Track is divided into evenly sized sectors



Optical Discs

A **CD-ROM** can be read from but not written to.

A **CD-R** is an optical disc on which users can write once, but not erase

A **CD-RW** is an erasable multisession disc

CD Versus DVD

- DVDs more storage capacity (up to 17 GB) than CDs (about 700 MB)
- More advanced laser reading system.
- DVDs to store video, while CDs are typically used for audio and smaller software programs.

Optical Discs

A **DVD-ROM** is a high-capacity optical disc on which users can read but not write on or erase

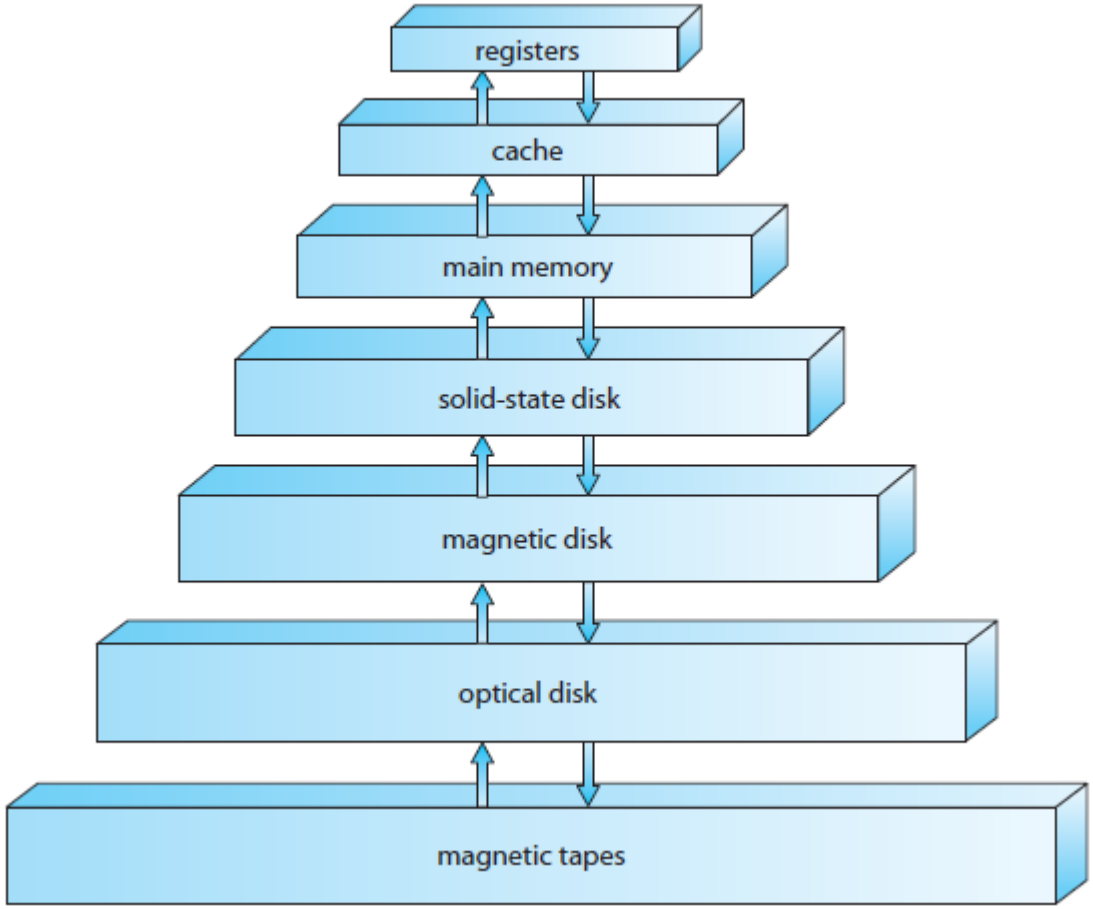
A **DVD-R** or **DVD+R** are competing DVD-recordable WORM formats, on which users can write once but not erase

DVD-RW, **DVD+RW**, and **DVD+RAM** are competing DVD-rewritable formats that users can write on multiple times

Magnetic Tape

- Tape is a magnetically coated ribbon of plastic that is capable of storing large amounts of data at a low cost.
- Tape no longer is used as a primary method of storage.
- Businesses use tape most often for long-term storage and backup.

Storage Hierarchy



Reference/Self Study

- Ch-8